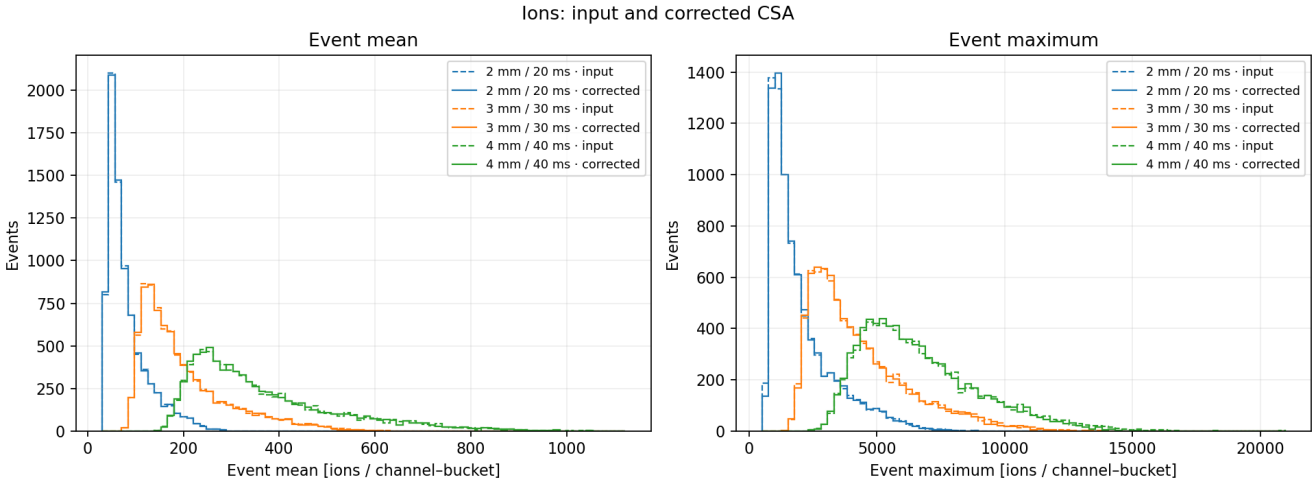
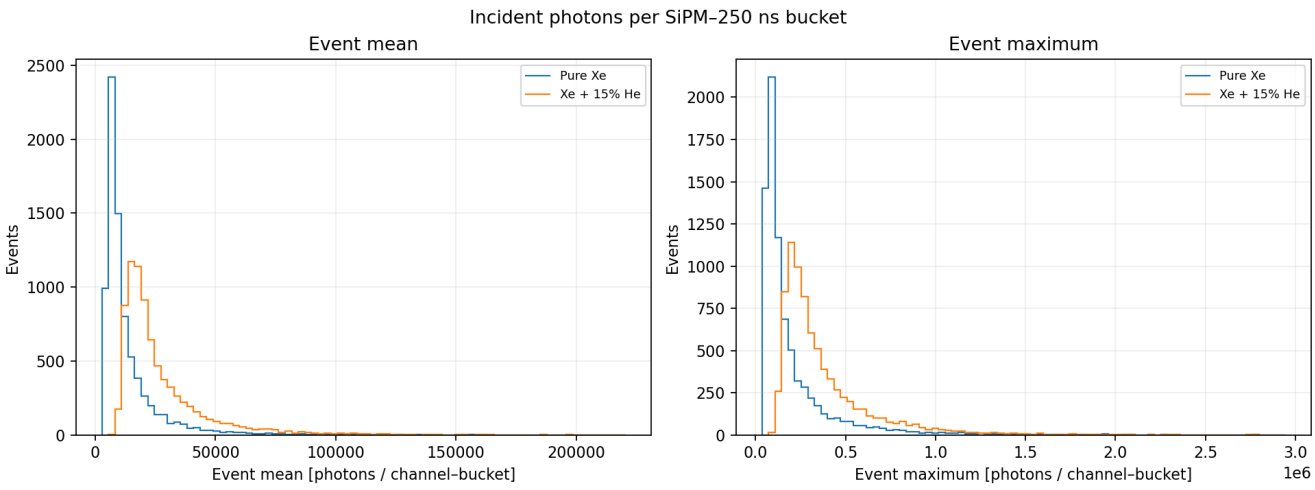
Ion and electron response: bb0nu

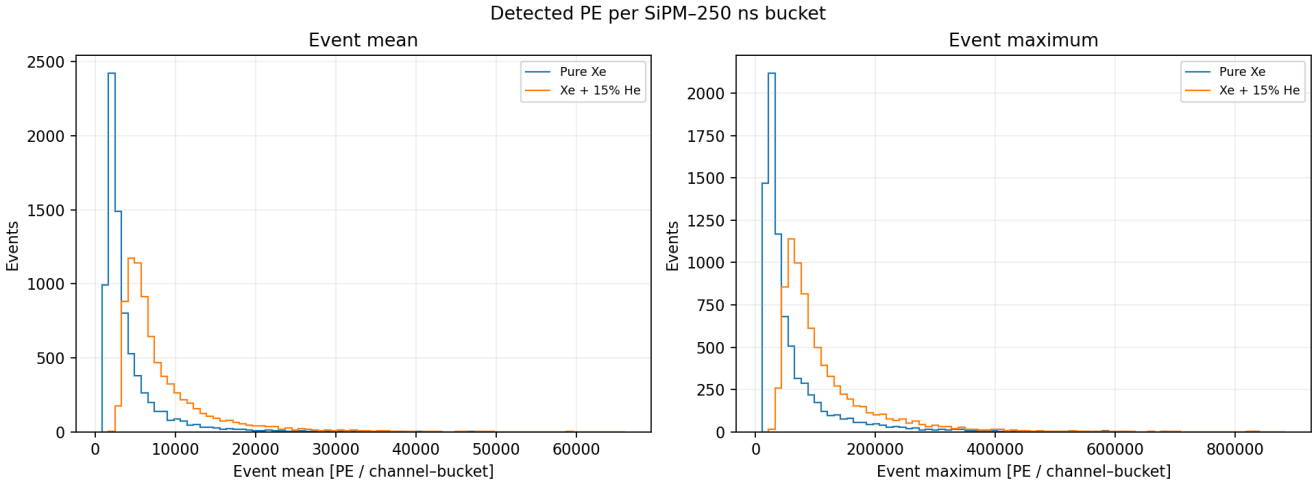
NAUSICAA

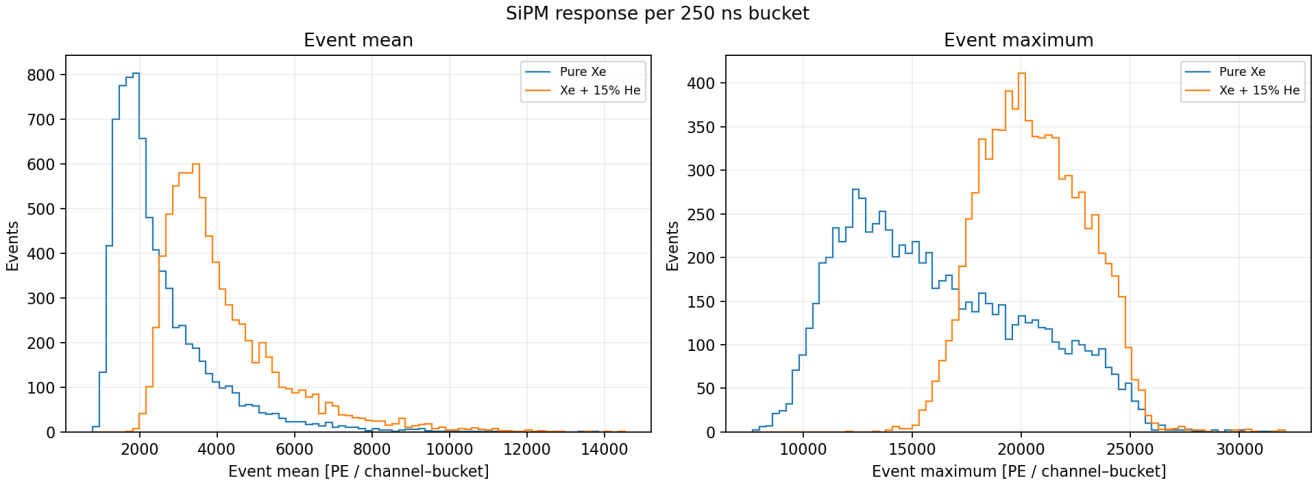
8,000 paired bb0nu events. Each event contributes one mean and one maximum. Ions use pixels with positive input ions; electrons use SiPM buckets with positive incident photons. The same cells enter all response stages, including signed reconstructed values. No amplitude or signal-to-noise cut is applied. Ion pitches 2, 3 and 4 mm use 20, 30 and 40 ms frames. Both electron gases use 10 mm pitch and 250 ns buckets. CSA results use the corrected reconstruction and the production 0.5 ENC modelling tolerance. These pulse summaries describe the occupied input support; event energy requires the complete response history.

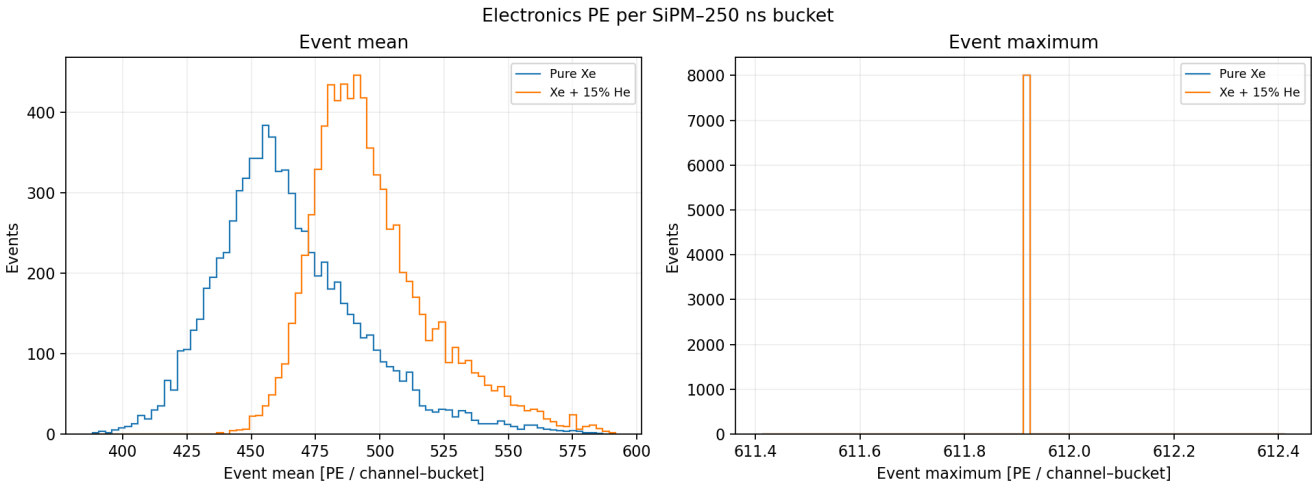
Stage	Events	Unit	Mean avg.	Mean median	Max avg.	Max median
ions 2mm corrected	8000	ions	81.870	67.147	2030.484	1572.565
ions 2mm input	8000	ions	81.890	67.211	2027.444	1570.000
ions 3mm corrected	8000	ions	199.811	170.505	4303.872	3766.422
ions 3mm input	8000	ions	200.254	170.582	4342.675	3787.000
ions 4mm corrected	8000	ions	364.472	320.348	6593.963	6159.345
ions 4mm input	8000	ions	367.585	321.506	6759.597	6245.000
pure xe detected	8000	PE	4183.219	2737.860	62512.329	35683.500
pure xe photons	8000	photons	13944.032	9124.571	208350.437	118787.000
pure xe readout	8000	PE	464.249	460.231	611.913	611.913
pure xe sensor	8000	PE	2516.532	2090.253	16132.001	15426.476
xe he15 detected	8000	PE	8320.653	6267.466	114773.285	85629.000
xe he15 photons	8000	photons	27735.572	20890.062	382535.649	285552.000
xe he15 readout	8000	PE	497.326	492.655	611.913	611.913
xe he15 sensor	8000	PE	4135.184	3677.519	20613.508	20466.305






Detected PE per SiPM–250 ns bucket
Event mean
Event maximum
Pure Xe
Xe + 15% He
Events
Event mean [PE / channel–bucket]
Event maximum [PE / channel–bucket]


SiPM response per 250 ns bucket
Event mean
Event maximum
Pure Xe
Xe + 15% He
Events
Event mean [PE / channel–bucket]
Event maximum [PE / channel–bucket]


Electronics PE per SiPM–250 ns bucket
Event mean
Event maximum
Pure Xe
Xe + 15% He
Events
Event mean [PE / channel–bucket]
Event maximum [PE / channel–bucket]

Reproduction

Numerical reduction: `scripts/production/reduce_response_overview.jl`.

Presentation: `scripts/production/present_response_overview.py`. The HDF5 reduction retains the source card, SiPM manifest, event IDs and reducer digest.